

Highlights

The first-passage time density for the diffusion model with variable drift and variable starting point

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- Closed-form density for the diffusion model with variable drift and starting point
- No closed form exists in the large-time series; the small-time series has one
- Analytic gradients; the seven-parameter likelihood needs one 1-D quadrature
- Verified symbolically, to 48 digits, against three packages, and in Lean 4
- Cheaper than every current package at the settings it fits with

The first-passage time density for the diffusion model with variable drift and variable starting point

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Abstract

The Ratcliff diffusion model is fitted with across-trial variability in drift rate, starting point, and non-decision time. The drift integral of the first-passage time density has been available in closed form; the starting-point integral has been evaluated numerically in every implementation, and Tuerlinckx (2004) concluded that no closed form exists. We show that this holds only for the large-time series representation. In the small-time representation the starting-point integral is a Gaussian moment, and we give the density with normally distributed drift and uniformly distributed starting point in closed form, together with a large-time counterpart in terms of the Faddeeva function. All partial derivatives are closed form, and the seven-parameter likelihood needs a single one-dimensional quadrature. The result is verified symbolically, to 48 digits, against three reference implementations and process simulation, and is machine-checked in Lean 4.

Keywords: Diffusion model, Response time modeling, First-passage time, Starting point variability

1. Background

The diffusion model for response times (Ratcliff, 1978) describes a two-choice decision as a Wiener process with drift v and unit diffusion coefficient¹

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¹For diffusion coefficient $\sigma^2 \neq 1$ all results below hold after the substitution $(v, \eta^2, a) \rightarrow (v/\sigma, \eta^2/\sigma^2, a/\sigma)$ (Blurton et al., 2017).

4 evolving between two absorbing barriers at 0 and $a > 0$ from a starting point
 5 aw , $0 < w < 1$. The first-passage times at the two barriers are the predicted
 6 response times for the two alternatives. To account for the fast and slow
 7 errors observed in most response time experiments, the model is applied
 8 with across-trial variability in three of its quantities (Ratcliff, 1978; Ratcliff
 9 and Rouder, 1998; Ratcliff and Tuerlinckx, 2002): the drift rate is normally
 10 distributed, $v \sim N(\nu, \eta^2)$; the starting point is uniformly distributed, $w \sim$
 11 $U(w_1, w_2)$, with Ratcliff's $s_z = a(w_2 - w_1)$; and the non-decision time is
 12 uniformly distributed over a window of width s_t . The predicted density of a
 13 response time is then a triple integral of the constant-parameter first-passage
 14 time density over these three distributions.

15 Of the three integrals, only the drift integral has been available in closed
 16 form. The first-passage time density with normally distributed drift is given
 17 by Horrocks and Thompson (2004) and, for the lower barrier, reads (Blurton
 18 et al., 2017, Eq. 1)

$$\begin{aligned}
 g(t \mid \nu, \eta^2, a, w) &= \frac{1}{\sqrt{2\pi t^3 S}} \exp\left[\frac{-\nu^2 t - 2\nu aw + \eta^2(aw)^2}{2S}\right] \\
 &\quad \times \sum_{j=0}^{\infty} (-1)^j r_j \exp\left(-\frac{r_j^2}{2t}\right), \tag{1}
 \end{aligned}$$

19 where $S := 1 + \eta^2 t$, $r_j = ja + aw$ for even j and $r_j = ja + a(1 - w)$ for odd
 20 j . Blurton et al. (2017) derived the corresponding cumulative distribution
 21 function. Both results are stated in the small-time representation of the first-
 22 passage time series (Navarro and Fuss, 2009; Gondan et al., 2014), that is,
 23 as a sum over image charges r_j with Gaussian factors $\exp(-r_j^2/2t)$.

24 The starting-point and non-decision-time integrals have remained numer-
 25 ical. Tuerlinckx (2004, pp. 706–708) worked in the large-time representation
 26 of Ratcliff (1978), in which the starting point enters through $\sin(k\pi w)$, inte-
 27 grated the starting point analytically at fixed drift, and concluded that the
 28 drift integral then has to be approximated by quadrature. Every current
 29 implementation of the seven-parameter model integrates the starting point
 30 numerically: DMAT (Vandekerckhove and Tuerlinckx, 2008), fast-dm and
 31 rtdists (Voss and Voss, 2007; Singmann et al., 2022), HDDM (Wiecki et al.,
 32 2013), WienR (Hartmann and Klauer, 2021), and the Stan implementation
 33 of Henrich et al. (2024). Blurton et al. (2017) noted that adding starting-
 34 point and non-decision-time variability to their result “requires numerical
 35 evaluation of two integrals” (p. 11).

36 In this note we show that the starting-point integral of the density (1)
 37 has a closed form. The reason it was missed is representational. In the
 38 large-time series, after the drift has been integrated, the starting point ap-
 39 pears in a Gaussian factor with a *positive* coefficient of w^2 and the integral
 40 against $\sin(k\pi w)$ lands on the imaginary error function; this is the obstruc-
 41 tion Tuerlinckx encountered. In the small-time series the factor $\exp(-r_j^2/2t)$
 42 contributes a negative coefficient of w^2 that always dominates the positive
 43 one produced by the drift integral, so the integrand is a linear function times
 44 a decaying Gaussian, and the integral is a Gaussian moment with a closed
 45 form. Section 2 gives the result and its derivation, Section 3 the large-time
 46 counterpart that is needed for numerical stability at large t , Section 4 the
 47 truncation rule, Section 5 the partial derivatives, Section 6 the resulting
 48 seven-parameter likelihood, and Section 7 the numerical verification and a
 49 comparison of computational cost with existing software.

50 **2. The density with variable drift and variable starting point**

51 Let the starting point be uniformly distributed on $[w_1, w_2]$, $0 \leq w_1 <$
 52 $w_2 \leq 1$. The lower-barrier first-passage time density of the resulting process
 53 is

$$g(t \mid \nu, \eta^2, a, w_1, w_2) := \frac{1}{w_2 - w_1} \int_{w_1}^{w_2} g(t \mid \nu, \eta^2, a, w) dw. \quad (2)$$

54 Because $r_j \geq ja$ for $w \in [0, 1]$, the j -th term of (1) is bounded in absolute
 55 value by $(j+1)a \exp(-j^2 a^2/2t)$ times a constant independent of w and j
 56 (Appendix B), so the series converges absolutely and uniformly on $[w_1, w_2]$
 57 and summation and integration may be exchanged. It therefore suffices to
 58 integrate one term.

59 *2.1. The j -th term is a Gaussian moment*

60 Write $r_j = p_j + q_j w$ with $p_j = ja$, $q_j = a$ for even j , and $p_j = (j+1)a$,
 61 $q_j = -a$ for odd j . Collecting the w -dependence of the j -th term of (1), that
 62 is, of $\exp[(-2\nu aw + \eta^2 a^2 w^2)/2S] r_j \exp(-r_j^2/2t)$, gives a quadratic exponent
 63 in w ,

$$(p_j + q_j w) \exp\left(-\frac{1}{2}Aw^2 + B_j w + C_j\right), \quad (3)$$

64 with

$$A = \frac{a^2}{tS}, \quad B_j = -\frac{\nu a}{S} \mp \frac{k_j a^2}{t}, \quad C_j = -\frac{\nu^2 t}{2S} - \frac{k_j^2 a^2}{2t}, \quad (4)$$

where $k_j = j$ and the upper sign apply for even j , and $k_j = j + 1$ and the lower sign apply for odd j . The coefficient of w^2 deserves attention. The drift integral contributes $+\eta^2 a^2/2S$, a growing Gaussian in w , whereas the image-charge factor contributes $-a^2/2t$, and

$$\frac{a^2}{2t} - \frac{\eta^2 a^2}{2S} = \frac{a^2(S - \eta^2 t)}{2tS} = \frac{a^2}{2tS} > 0 \quad (5)$$

for every $t > 0$, every η , and every j . The combined exponent is therefore a decaying Gaussian with $A > 0$. This is the whole content of the result: in the small-time representation the starting-point integrand is a linear function times a proper Gaussian, and such integrals have closed forms in terms of $\exp$ and Φ .

2.2. The result

For $A > 0$ and $m := B/A$,

$$\begin{aligned} & \int_{w_1}^{w_2} (p + qw) \exp(-\tfrac{1}{2}Aw^2 + Bw + C) dw \\ &= (p + qm) \sqrt{\frac{2\pi}{A}} \exp(C + B^2/2A) \left[\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 - m)) \right] \\ & - \frac{q}{A} \left[\exp(C + Bw_2 - \tfrac{1}{2}Aw_2^2) - \exp(C + Bw_1 - \tfrac{1}{2}Aw_1^2) \right], \end{aligned} \quad (6)$$

where Φ denotes the standard normal distribution function (Appendix A). Applying (6) term by term with the coefficients (4) gives the lower-barrier first-passage time density with normally distributed drift and uniformly distributed starting point,

$$\begin{aligned} g(t \mid \nu, \eta^2, a, w_1, w_2) &= \frac{1}{(w_2 - w_1)\sqrt{2\pi t^3 S}} \sum_{j=0}^{\infty} (-1)^j I_j, \\ I_j &= \int_{w_1}^{w_2} (p_j + q_j w) \exp(-\tfrac{1}{2}Aw^2 + B_j w + C_j) dw, \end{aligned} \quad (7)$$

with I_j given in closed form by (6). The density at the upper barrier is $g(t \mid -\nu, \eta^2, a, 1 - w_2, 1 - w_1)$. For $w_2 - w_1 \rightarrow 0$ the result reduces to (1), and for $\eta = 0$ it reduces to the starting-point integral of the constant-drift density, so it can be used in a fitting routine regardless of whether either form of variability is present in the data.

Two remarks on evaluation. First, for large j the center $m = B_j/A$ of the Gaussian in (6) lies far outside $[w_1, w_2]$, so that $\exp(C + B^2/2A)$ is very large and the bracketed difference of Φ values is very small. The product must be formed in logarithmic space, using $\log \Phi$ and a tail-aware difference; with that precaution the expression is stable for all parameter values we have tested. Second, the identity $\sqrt{2\pi} \exp(C + B^2/2A) \phi(\sqrt{A}(w_i - m)) = \exp(C + Bw_i - \frac{1}{2}Aw_i^2)$, where ϕ is the standard normal density, shows that the two brackets in (6) share their exponential factors; this is used in the derivatives of Section 5.

3. Large-time representation

Like every small-time series, (7) loses precision through cancellation when t/a^2 is large; in double precision the loss becomes visible for $t/a^2 \gtrsim 2$. A large-time counterpart is obtained from the large-time series of Ratcliff (1978). Integrating $\exp(-vaw - v^2t/2)$ against the $N(\nu, \eta^2)$ drift density produces exactly the prefactor of (1), so the large-time density with normally distributed drift is

$$g(t \mid \nu, \eta^2, a, w) = \frac{\pi}{a^2 \sqrt{S}} \exp \left[\frac{-\nu^2 t - 2\nu a w + \eta^2 (aw)^2}{2S} \right] \times \sum_{k=1}^{\infty} k \exp(-k^2 \pi^2 t / 2a^2) \sin(k\pi w). \quad (8)$$

Here there is no image-charge factor to compensate the growing Gaussian, and the starting-point integral of the k -th term is $\int_{w_1}^{w_2} \exp(\kappa w^2 + \lambda w) \sin(k\pi w) dw$ with $\kappa = \eta^2 a^2 / 2S > 0$ and $\lambda = -\nu a / S$. Written as the imaginary part of $\int \exp(\kappa w^2 + \mu_k w) dw$ with $\mu_k = \lambda + ik\pi$, the antiderivative is an imaginary error function of complex argument, which overflows. It is, however, bounded when expressed through the Faddeeva function $w(z) = \exp(-z^2) \operatorname{erfc}(-iz)$ (Abramowitz and Stegun, 1964, 7.1.3):

$$\int_{w_1}^{w_2} \exp(\kappa w^2 + \mu_k w) dw = \frac{-i\sqrt{\pi}}{2\sqrt{\kappa}} \left[\exp(\kappa w^2 + \mu_k w) w(z_k(w)) \right]_{w_1}^{w_2}, \quad (9)$$

with $z_k(w) = \sqrt{\kappa} w + \mu_k / 2\sqrt{\kappa}$. Since $\operatorname{Im} z_k = k\pi / 2\sqrt{\kappa} > 0$, $w(z_k)$ lies in the upper half plane where it is bounded by one, and the product with

110 $\exp(\kappa w^2 + \mu_k w)$ is of the size of the integrand. For $\eta = 0$ the bracket re-
 111 duces to $(\exp(\mu_k w_2) - \exp(\mu_k w_1))/\mu_k$. The large-time density with uniformly
 112 distributed starting point is then

$$g(t \mid \nu, \eta^2, a, w_1, w_2) = \frac{\pi \exp(-\nu^2 t / 2S)}{a^2 \sqrt{S} (w_2 - w_1)} \sum_{k=1}^{\infty} k \exp(-k^2 \pi^2 t / 2a^2) \\ \times \text{Im} \int_{w_1}^{w_2} \exp(\kappa w^2 + \mu_k w) dw. \quad (10)$$

113 Fast and accurate algorithms for $w(z)$ are standard (Zaghloul and Ali, 2011)
 114 and available in scientific libraries. In our implementation the two represen-
 115 tations are switched at $t/a^2 = 1$; they agree to 1.5×10^{-11} relative across
 116 the switch. We stress that (9) must be used in the Faddeeva form: written
 117 with the imaginary error function it loses of the order of $\pi^2 k^2 / 4\kappa$ nats to
 118 cancellation and returns meaningless values when the series is truncated at
 119 a fixed large K .

120 4. Convergence

121 The j -th term of (7) is bounded in absolute value by $c(j +$
 122 $1)a \exp(-j^2 a^2 / 2t)$, and the k -th term of the large-time series (10) by
 123 $c' k \exp(-k^2 \pi^2 t / 2a^2)$, where the constants c and c' do not depend on the
 124 index (Appendix B). Both bounds decrease super-exponentially, so the trun-
 125 cation error is of the order of the first neglected term, and for a tolerance ε
 126 the series may be cut at

$$J = \left\lceil \frac{\sqrt{2t \log(1/\varepsilon)}}{a} \right\rceil + 1, \quad K = \left\lceil a \sqrt{\frac{2 \log(1/\varepsilon)}{\pi^2 t}} \right\rceil + 1. \quad (11)$$

127 With $\varepsilon = 10^{-12}$ the small-time series needs $J \leq 9$ terms for $t/a^2 \leq 1$, and
 128 the large-time series needs $K \leq 4$ terms for $t/a^2 \geq 1$. Evaluating (7) with
 129 $\varepsilon = 10^{-12}$ and with $\varepsilon = 10^{-40}$ on the 400 parameter sets of Section 7 gives
 130 results that differ by at most 2.2×10^{-16} .

131 5. Partial derivatives

132 Gradient information is required by Hamiltonian Monte Carlo and by
 133 gradient-based optimizers, and Hartmann and Klauer (2021) derived it for

134 the constant-parameter model. Every term of (7) and (10) is a composition
 135 of $\exp$, Φ , and w , so the partial derivatives with respect to ν , η , a , w_1 , and
 136 w_2 are again closed form. For the small-time representation it is convenient
 137 to differentiate I_j with respect to its seven coefficients and to apply the chain
 138 rule. Writing $G := \sqrt{2\pi/A} \exp(C + B^2/2A)[\Phi(\sqrt{A}(w_2 - m)) - \Phi(\sqrt{A}(w_1 -$
 139 $m))]$ and $E_i := \exp(C + Bw_i - \frac{1}{2}Aw_i^2)$, so that $I = (p + qm)G - (q/A)(E_2 - E_1)$,
 140 the total differential of I under $(dA, dB, dC, dp, dq, dw_1, dw_2)$ is

$$\begin{aligned}
 dI &= (dp + m dq + q dm) G + (p + qm) dG \\
 &\quad - \left(\frac{dq}{A} - \frac{q dA}{A^2} \right) (E_2 - E_1) - \frac{q}{A} (dE_2 - dE_1), \\
 dm &= \frac{dB - m dA}{A}, \quad dK = dC + m dB - \frac{1}{2}m^2 dA, \\
 dG &= G \left(dK - \frac{dA}{2A} \right) + E_2 \left[(dw_2 - dm) + \frac{(w_2 - m) dA}{2A} \right] \\
 &\quad - E_1 \left[(dw_1 - dm) + \frac{(w_1 - m) dA}{2A} \right], \\
 dE_i &= E_i \left[dC + w_i dB - \frac{1}{2}w_i^2 dA + (B - Aw_i) dw_i \right], \tag{12}
 \end{aligned}$$

141 in which the identity noted at the end of Section 2 has been used to express
 142 dG through E_1 and E_2 rather than through ϕ . The coefficient differentials
 143 follow from (4); for instance $\partial B_j / \partial \nu = -a/S$ and $\partial C_j / \partial \nu = -\nu t/S$, and
 144 A , p_j , q_j do not depend on ν . For the large-time representation only two
 145 identities are needed, both immediate from $\frac{d}{dw} \exp(\kappa w^2 + \mu w) = (2\kappa w +$
 146 $\mu) \exp(\kappa w^2 + \mu w)$: with $\mathcal{I} := \int_{w_1}^{w_2} \exp(\kappa w^2 + \mu w) dw$ and $\mathcal{E}_i := \exp(\kappa w_i^2 + \mu w_i)$,

$$\frac{\partial \mathcal{I}}{\partial \mu} = \frac{(\mathcal{E}_2 - \mathcal{E}_1) - \mu \mathcal{I}}{2\kappa}, \quad \frac{\partial \mathcal{I}}{\partial \kappa} = \frac{(w_2 \mathcal{E}_2 - w_1 \mathcal{E}_1) - \mathcal{I} - \mu \partial \mathcal{I} / \partial \mu}{2\kappa}. \tag{13}$$

147 The derivatives with respect to w_1 and w_2 are, by the fundamental theorem
 148 of calculus, the integrand evaluated at the endpoints, corrected for the fac-
 149 tor $1/(w_2 - w_1)$. All of (12) and (13), and every chain-rule coefficient, are
 150 verified as exact symbolic identities in the supplementary code; the resulting
 151 gradients agree with those of Hartmann and Klauer (2021) as implemented
 152 in WienR to 3×10^{-13} absolute on the parameter grid of Section 7.

Table 1: Maximum relative error of Gauss–Legendre quadrature for (14) against adaptive quadrature, 200 random parameter sets, densities above 10^{-3} .

Nodes	Plain	With cubic map
8	4.1×10^{-3}	4.3×10^{-3}
16	7.3×10^{-5}	3.5×10^{-6}
24	4.1×10^{-6}	2.7×10^{-8}
32	1.1×10^{-8}	1.5×10^{-9}

153 6. The seven-parameter likelihood

154 With the non-decision time uniformly distributed on $[t_0, t_0 + s_t]$, the den-
155 sity of an observed response time is

$$f(t) = \frac{1}{s_t} \int_{t-t_0-s_t}^{t-t_0} g(u \mid \nu, \eta^2, a, w_1, w_2) du, \quad (14)$$

156 where the lower limit is replaced by 0 if negative. This integral does not
157 have a closed form: the non-decision time shifts the time argument, which
158 enters every term of (7) through $\sqrt{t}$ and $1/t$, and we have not found a
159 representation in which the shift integrates. What (7) achieves is to reduce
160 the seven-parameter likelihood from a two-dimensional cubature over (w, u)
161 to a one-dimensional quadrature over u of a smooth closed-form integrand
162 on a short interval. Because $g(u)$ rises steeply from zero at $u = 0$ and
163 the window may include that leading edge, a plain Gauss–Legendre rule
164 converges slowly; the substitution $u = u_{lo} + (u_{hi} - u_{lo}) s^3$, which clusters
165 nodes at the lower limit, fixes this. Table 1 gives the maximum relative error
166 of the rule against adaptive quadrature over the 200 seven-parameter sets
167 described below. A 32-node rule with the cubic map reaches 1.5×10^{-9} at
168 a fixed cost of 32 evaluations of (7). The gradient of (14) with respect to
169 the five parameters of g is the same rule applied to the gradient of g , and
170 $\partial f / \partial t_0 = [g(u_{lo}) - g(u_{hi})] / s_t$ and $\partial f / \partial s_t = [g(u_{lo}) - f] / s_t$ follow from the
171 endpoints.

172 7. Verification and efficiency

173 7.1. Verification

174 We checked the result in five independent ways. (i) Every algebraic step,
175 in particular the antiderivative (6), the inequality (5), the coefficient ta-

176 ble (4), the drift integral leading to (1), and all derivative formulae of Sec-
 177 tion 5, is verified as an exact symbolic identity in a computer algebra system,
 178 that is, as $\text{simplify}(\text{d}F/\text{d}x - f) = 0$ rather than by numerical agreement.
 179 (ii) In 40- to 60-digit arithmetic, (7) agrees with a brute-force double integral
 180 of the constant-parameter density over (v, w) to 8.7×10^{-32} , and (10) agrees
 181 with quadrature of (1) to between 10^{-35} and 10^{-48} at $t/a^2 = 0.9, 1.5$, and 4.7 .
 182 (iii) On 400 random parameter sets with $a \in \{0.6, 1, 1.5, 2.5\}$, $\nu \in [-3, 4]$,
 183 $\eta \in \{0.5, 1, 2\}$, $w_2 - w_1 \in \{0.05, 0.2, 0.4\}$, and $t \in [0.05, 3]$, spanning densi-
 184 ties from 5×10^{-20} to 1.6 , the switched representation agrees with WienR
 185 (Hartmann and Klauer, 2021) called at its tightest tolerance (10^{-12}) to a
 186 maximum absolute difference of 1.2×10^{-14} ; `rtdists` (Singmann et al., 2022),
 187 whose diffusion code is that of `fast-dm`, converges onto the same values as
 188 its precision setting is raised (7.6×10^{-4} , 1.1×10^{-5} , 2.5×10^{-8} at precision
 189 3, 5, 8), as does the HDDM likelihood (Wiecki et al., 2013) as its integra-
 190 tion tolerance is tightened. On 200 seven-parameter sets, (14) agrees with
 191 WienR to 2.5×10^{-9} , which is WienR’s own reported cubature error. (iv) An
 192 Euler–Maruyama simulation of the process with 400 000 trials, step 5×10^{-5} ,
 193 and the Broadie–Glasserman–Kou barrier correction gives a lower-barrier re-
 194 sponse time histogram in 30 bins of 100 ms whose χ^2 against the binned
 195 density (7) is 21.6, with all standardized residuals below 1.9. (v) The deriva-
 196 tion of (7), including the antiderivative, the coefficient table, the exchange of
 197 summation and integration, and the Faddeeva integral (9), is formalized and
 198 machine-checked in Lean 4 with Mathlib (de Moura and Ullrich, 2021; The
 199 mathlib Community, 2020); the development contains no unproved state-
 200 ment and relies on no axioms beyond Lean’s standard ones. The starting
 201 representations (1) and (8) are taken as definitions there, as they are taken
 202 from the literature here.

203 7.2. Efficiency

204 Table 2 compares the cost of (7) and (14), compiled from a direct tran-
 205 scription of the formulae, with the three packages in common use: WienR,
 206 which evaluates the drift-integrated series by adaptive cubature over w ; `rt-`
 207 `dists`, whose diffusion code is that of `fast-dm` and integrates over w and the
 208 non-decision window by adaptive Simpson’s rule; and the HDDM likelihood,
 209 which does the same with the Navarro–Fuss series. All timings are single-

Table 2: Cost per evaluation in microseconds, single thread, 10 000 response times at $a = 1.2$, $\nu = 1$, $w = 0.5$, $\eta = 1$, $w_2 - w_1 = 0.2$, $s_t = 0.2$. Each package at the settings its fitting routine uses: WienR at tolerance 10^{-12} , rtdists at precision 3, HDDM at series tolerance 10^{-4} with Simpson tolerance 10^{-8} and recursion depth 10. Error is the maximum relative deviation from (7) or (14) over the parameter sets of Section 7 (densities above 10^{-3}); for the closed form the seven-parameter entry is the quadrature error of (14) against adaptive integration.

	Six parameters		Seven parameters (+ s_t)	
	Error	μs	Error	μs
Closed form	–	0.33	1.5×10^{-9}	9.2
WienR	2.5×10^{-12}	2.1	1.5×10^{-9}	1040
rtdists	4.6×10^{-3}	0.43	9.2×10^{-3}	18
HDDM	3.9×10^{-6}	0.60	1.5×10^{-7}	46
Closed form + 5 gradients [†]	–	1.7		
WienR, 5 gradients	10^{-13}	15.7		

[†]Uncompiled vectorized implementation.

210 threaded on one machine,² for 10 000 response times at one parameter set,
 211 which is the situation in a likelihood evaluation.

212 The error columns need a reference scale. Blurton et al. (2017) recom-
 213 mend a truncation tolerance of 1.5×10^{-8} , the square root of single-precision
 214 machine epsilon, and Gondan et al. (2014) illustrate their term counts at
 215 10^{-3} . Tuerlinckx (2004, Table 1) reports differences of 2×10^{-7} to 3×10^{-4}
 216 between his quadrature-based algorithm and the classical one. Among cur-
 217 rent packages, rtdists (fast-dm) defaults to a precision of 3, documented as
 218 roughly the number of accurate decimals; the HDDM likelihood truncates
 219 its series at 10^{-4} and integrates over the starting point and the non-decision
 220 window by adaptive Simpson’s rule with tolerance 10^{-8} and recursion depth
 221 10; the Stan implementation of Henrich et al. (2024) fixes the density tol-
 222 erance at 10^{-6} and defaults to 10^{-4} for the derivatives; WienR defaults to
 223 10^{-12} . The closed form is exact to double precision, and its seven-parameter
 224 quadrature at 1.5×10^{-9} is below every one of these tolerances.

225 Two features of Table 2 are worth stating. First, the closed form is the

²Apple M1 Pro; R 4.5.3 with WienR 0.3-15 and rtdists 0.11-6; Python 3.11 with NumPy 1.26.4, SciPy 1.13.1, Cython 3.1.4, and hddm-wfpt 0.1.4, the HDDM likelihood distributed as a standalone package.

226 cheapest of the four at every entry and the only one that is exact for six
 227 parameters. `rtdists` costs 1.3 times as much for six parameters and twice as
 228 much for seven, and deviates from the density by 0.5% and 0.9%. HDDM
 229 costs 1.8 and 5 times as much and agrees to 4×10^{-6} and 1.5×10^{-7} ; its
 230 series tolerance of 10^{-4} leaves a floor near 10^{-7} that tightening its integration
 231 settings does not remove. WienR reaches 10^{-12} at 6 and 110 times the cost.
 232 Because the cost of the closed form is fixed whereas that of the integrators
 233 is set by the accuracy asked of them, these ratios grow with the tolerance.
 234 Second, the gradient comes almost free: all five partial derivatives of (7) share
 235 the exponential factors of the density itself, and an uncompiled vectorized
 236 implementation returns the density together with all five derivatives in $1.7 \mu\text{s}$,
 237 whereas WienR requires one further cubature per derivative. For reference,
 238 WienR’s own closed form for drift variability alone (Blurton et al., 2017)
 239 evaluates in $0.26 \mu\text{s}$ on the same machine, so the closed form with starting-
 240 point variability costs 1.3 times the drift-only density.

241 8. Discussion

242 We have shown that the first-passage time density of the diffusion model
 243 with normally distributed drift and uniformly distributed starting point has a
 244 closed form, and that the reason it was not found earlier is the choice of series
 245 representation: the small-time series carries a Gaussian factor in the starting
 246 point that the large-time series lacks, and that factor is exactly what gives
 247 the integral a closed form. The result completes the analytic treatment of
 248 the drift and starting-point integrals of the seven-parameter model and leaves
 249 only the non-decision-time integral, which is a one-dimensional quadrature
 250 of a smooth function on a short interval and for which we have given a fixed
 251 32-node rule accurate to 10^{-9} .

252 For practice the consequences are threefold. The density is exact up to
 253 a truncation error that is controlled by (11); it is evaluated at a fixed cost
 254 that is below that of each existing package at the settings the package fits
 255 with; and its partial derivatives are closed form and nearly free, which is
 256 what hierarchical Bayesian and Hamiltonian Monte Carlo implementations
 257 (Wiecki et al., 2013; Henrich et al., 2024) need. The result is directly usable
 258 in any of the existing packages, which already contain the drift-integrated
 259 density (1) as a building block and need only replace the quadrature over w
 260 by (6).

261 The corresponding cumulative distribution function with both forms of
262 variability does not follow by the same route. Starting from the drift-
263 integrated distribution function of Blurton et al. (2017), half of the series
264 terms have a coefficient of w^2 that is exactly zero after the drift integral and
265 close through the integral $\int \exp(cw)\Phi(\alpha w + \beta) dw$, which has a closed form;
266 the other half carry the coefficient $+2a^2\eta^2$, a growing Gaussian times Φ , and
267 reduce to an integral of Owen- T type that has no closed form in terms of
268 $\exp$, Φ , or Owen's T . For maximum likelihood and Bayesian estimation the
269 density suffices, and we leave the distribution function open.

270 CRediT authorship contribution statement

271 **Kiante Fernandez:** Conceptualization, Methodology, Software, Valida-
272 tion, Formal analysis, Writing – original draft, Writing – review & editing.

273 Declaration of competing interest

274 The author declares no competing interests.

275 Data availability

276 All code, reference values, comparison scripts, and the Lean 4 formaliza-
277 tion are available in the supplementary material and at [repository URL to
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282 Appendix A. The antiderivative

283 Let $A > 0$, $m = B/A$, and

$$F(w) := (p + qm)\sqrt{\frac{2\pi}{A}} \exp(C + B^2/2A) \Phi(\sqrt{A}(w - m)) \\ - \frac{q}{A} \exp(C + Bw - \frac{1}{2}Aw^2).$$

284 Using $\Phi' = \phi$ and $-\frac{1}{2}A(w - m)^2 + C + B^2/2A = C + Bw - \frac{1}{2}Aw^2$,

$$\begin{aligned} F'(w) &= (p + qm)\sqrt{2\pi} \exp(C + B^2/2A) \phi(\sqrt{A}(w - m)) \\ &\quad - \frac{q}{A}(B - Aw) \exp(C + Bw - \frac{1}{2}Aw^2) \\ &= [(p + qm) - q(m - w)] \exp(C + Bw - \frac{1}{2}Aw^2) \\ &= (p + qw) \exp(C + Bw - \frac{1}{2}Aw^2), \end{aligned}$$

285 which is the integrand of (6). The definite integral (6) is $F(w_2) - F(w_1)$.
 286 The same computation is carried out symbolically in the supplementary code
 287 and formally in the Lean development.

288 Appendix B. Term bounds

289 For $w \in [0, 1]$, $ja \leq r_j \leq (j + 1)a$, and the prefactor of (1) is bounded
 290 by $\exp[(2|\nu|a + \eta^2 a^2)/2S]$. Hence the j -th term of the series in (1), and
 291 therefore $|I_j|/(w_2 - w_1)$ in (7), is at most $(j + 1)a \exp(-j^2 a^2/2t)$ times a
 292 constant independent of j and w . Since $\sum_j (j + 1) \exp(-j^2 c)$ converges for
 293 every $c > 0$, the series converges absolutely and uniformly in w , which justifies
 294 the exchange of summation and integration and gives the truncation rule for
 295 J in (11). For the large-time series (10), $|w(z)| \leq 1$ in the upper half plane
 296 and $|\exp(\kappa w^2 + \mu_k w)| \leq \exp(\kappa + |\lambda|)$ on $[0, 1]$, so the k -th term is at most a
 297 constant times $k \exp(-k^2 \pi^2 t/2a^2)$, which gives the rule for K .

298 Appendix C. Supplementary material

299 The supplementary material contains (i) a vectorized implementation
 300 of (7), (10), (14), and all partial derivatives in Python, together with a
 301 compiled kernel used for Table 2; (ii) the symbolic verification script; (iii)
 302 the scripts and reference values for the comparisons with WienR, rtdists, and
 303 HDDM, and the simulation; and (iv) the Lean 4 formalization.

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